

CE 559: Biological Wastewater Treatment

Course Number: CE 559

Course Title: Biological Wastewater Treatment

L-T-P-C: 3-0-0-3

Prerequisites: None

Intended for: UG/M.Tech./MS/PhD

Distribution: Elective

Approval: 28th Senate

Course Contents

- **Introduction:** Need for wastewater treatment, Need for sludge treatment, Benefits of biological wastewater treatment, Overview of wastewater treatment processes. [2 Lectures]
- **Water and wastewater quality parameter:** pH and DO, Solids: total, dissolved, volatile and fixed solids, Organic carbon: BOD, COD and TOC, Nitrogen: Total Kjeldahl Nitrogen (TKN), Ammonia-nitrogen (NH₃N), Nitrite-nitrogen (NO₂N) and Nitrate-nitrogen (NO₃N), Phosphorous, Microbial analysis: Most probable number (MPN). [6 Lectures]
- **Process Analysis:** Rate of reaction, Order of reaction, Effect of temperature on reaction rate, Enzyme Reaction and kinetics, Types of reactors: CSTR and plug flow reactor, Reactor analysis, Residence time. [6 Lectures]
- **Biological Systems:** Overview and application in wastewater treatment, Biological growth and kinetics, Half-life and doubling time, Yield and decay, Monod's kinetics, Estimation of kinetic parameters. [6 Lectures]
- **Aerobic Process:** Suspended growth processes: Ponds and lagoons, Activated sludge process: Process description and its modifications, Substrate utilization and biomass growth, Process design, Hydraulic and Solids residence time, Activated sludge process with and without biomass recirculation, Efficiency and loading criteria, Effect of temperature on process performance, Trouble shooting, Attached growth processes: Rotating biological contactor (RBC), Trickling filter. [10 Lectures]
- **Anaerobic Processes:** Process description, Process design, Startup and operation, High-rate anaerobic processes, Biofilm and biofloc processes, Loading criteria and biogas generation rate, Biogas yield and composition, Biogas cleanup and use, Trouble shooting and maintenance, Sludge digestion. [6 Lectures]
- **Downstream Treatment:** Need for nutrient removal, Nitrogen removal processes, Biological nitrification and denitrification, Phosphorus removal, Disinfection. [6 Lectures]

Textbooks:

1. George Tchobanoglous, Franklin Burton, H. David Stensel, **Wastewater Engineering: Treatment and Reuse**, 4th Edition, McGraw-Hill International Edition, 2003.
2. Henze, M., Mark C. M. Van Loosdrecht, George A. Ekema, Damir Brdjanovic, **Biological Wastewater Treatment: Principles, Modelling and Design**, IWA Publishing, 1997.



Reference Books:

1. Sawyer, C. N., Parkin, G. F. and McCarty, P. L., **Chemistry for Environmental Engineering**, McGraw-Hill, 2008.
2. Bailey, J. E. and Ollis, D. F., **Biochemical Engineering Fundamentals**, 2nd Edition, McGraw-Hill International, 1987.